Practice workbook · Maximum or minimum? Using the second derivative

Showing a point is stationary and deciding whether it is a maximum or minimum from the sign of d2y/dx2.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-09

1. Question 7 · part c · 2 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

7 The curve C has equation $y = \dfrac{x}{x^2 + 4}$

(a) Using calculus, find the coordinates of the stationary points on C.

(5)

(b) Show that $\dfrac{d^2y}{dx^2} = \dfrac{2x(x^2 - 12)}{(x^2 + 4)^3}$

(4)

(c) Hence, or otherwise, determine the nature of each of these stationary points.

(2)

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1. Question 7 · part c · 2 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

Question 7 continued

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1. Question 7 · part c · 2 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

Question 7 continued

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1. Question 7 · part c · 2 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 11 marks)

1. Question 7 · part c · 2 marks · Working and answer

Working and answer

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2. Question 9 · part b.i · 0 marks

2019 JUN · Paper 02

Further practice · Attempt part b.i only.

9 The curve C, with equation $y = f(x)$, passes through the point with coordinates $\left(-2, -\frac{28}{3}\right)$

Given that $f'(x) = x^3 - x^2 - 4x + 4$

(a) show that C passes through the origin.

(4)

(b) (i) Show that C has a minimum point at $x = 2$ and a maximum point at $x = 1$

(ii) Find the exact value of the y coordinate at each of these points.

(7)

The curve has another turning point at A.

(c) (i) Find the coordinates of A.

(ii) Determine the nature of this turning point.

(3)

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2. Question 9 · part b.i · 0 marks · Working and answer

Working and answer

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3. Question 9 · part c.ii · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

9 The curve C, with equation $y = f(x)$, passes through the point with coordinates $\left(-2, -\frac{28}{3}\right)$

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3. Question 9 · part c.ii · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

Question 9 continued

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3. Question 9 · part c.ii · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

Question 9 continued

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3. Question 9 · part c.ii · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

Question 9 continued

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(Total for Question 9 is 14 marks)

3. Question 9 · part c.ii · 3 marks · Working and answer

Working and answer

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4. Question 7 · part b · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

7 The curve with equation $y = mx^2 + 64\sqrt{x} + 39$ has a stationary point with coordinates $(4, n)$ where m and n are integers.

Using calculus

(a) find the value of m and the value of n

(6)

(b) determine the nature of the stationary point.

(2)

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4. Question 7 · part b · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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Question 7 continued

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(Total for Question 7 is 8 marks)

4. Question 7 · part b · 2 marks · Working and answer

Working and answer

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5. Question 7 · part a.ii · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part a.ii only.

7 The point with coordinates $(4, -104)$ lies on the curve C with equation $y = f(x)$

Given that $f'(x) = 4x^3 - 12x^2 - 19x + 12$

(a) (i) show that C passes through the origin,

(4)

(ii) show that C has a maximum at the point on the curve where $x = 0.5$

(3)

The curve C has another turning point at A and another turning point at B.

Given that the x coordinate of A is negative,

(b) (i) find the coordinates of A and the coordinates of B,

(5)

(ii) determine the nature of these turning points.

(3)

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5. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

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6. Question 7 · part b.ii · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

7 The point with coordinates $(4, -104)$ lies on the curve C with equation $y = f(x)$

Given that $f'(x) = 4x^3 - 12x^2 - 19x + 12$

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Given that the x coordinate of A is negative,

(b) (i) find the coordinates of A and the coordinates of B,

(5)

(ii) determine the nature of these turning points.

(3)

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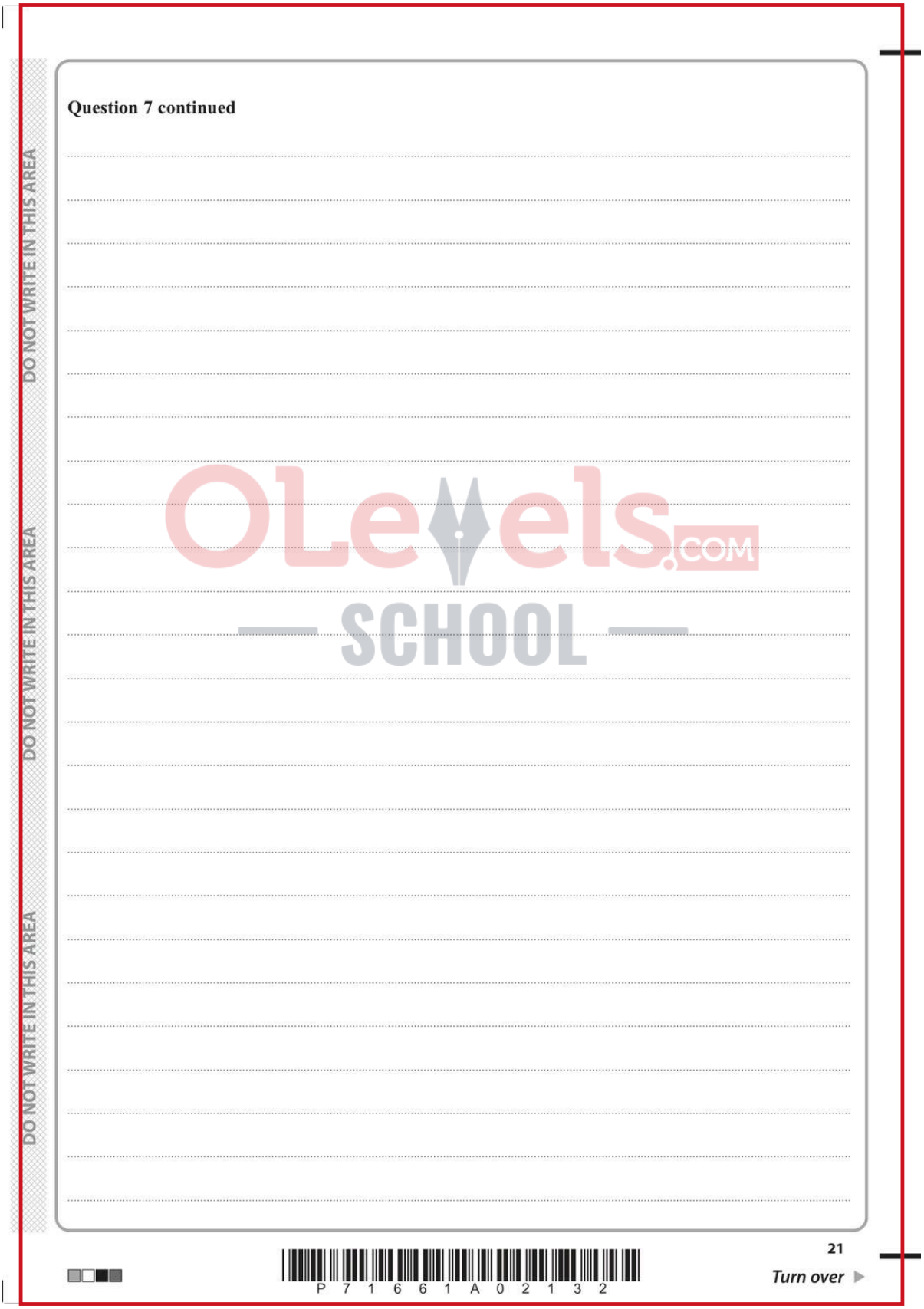
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6. Question 7 · part b.ii · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.



6. Question 7 · part b.ii · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

Question 7 continued

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6. Question 7 · part b.ii · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.



Question 7 continued

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(Total for Question 7 is 15 marks)

6. Question 7 · part b.ii · 3 marks · Working and answer

Working and answer

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7. Question 4 · part b · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part b only.

4 The curve C has equation $y = 2x^2 + px + q$ where p and q are integers.

The curve C has a stationary point at $(3, -5)$

(a) Show that $p = -12$ and find the value of q

(4)

(b) State, giving a reason, the nature of the stationary point.

(1)

(c) Find an equation of the normal to C at the point on C where $x = 1$

Give your answer in the form $ax + by + c = 0$

(6)

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7. Question 4 · part b · 1 mark · Working and answer

Working and answer

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8. Question 4 · part b · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

4 The curve C has equation

$$y = x^3 - 3x^2 - 24x + 6$$

(a) Use calculus to find the coordinates of each of the stationary points on C.

(4)

(b) Determine the nature of each of these stationary points.
Justify your answers.

(2)

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8. Question 4 · part b · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 6 marks)

8. Question 4 · part b · 2 marks · Working and answer

Working and answer

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